

Name: _____

Date: _____

Class: _____

Element K: Reflection on the Design Project

Engineering Design Process Summary (K1, K2)

Summarize each step of the design process including:

Initial meeting in TV studio:

The project started when Mr. O'Farrell gave us a briefing of the projects he needed to be worked on. Of the many options, our group chose the mic holder because we had a vision for how we would be able to solve the problem.

Creation of a design brief: Based on Mr. At O'Farrell's requests we created a design brief where we quickly analyzed previous designs and laid out functions, objectives, and constraints for this project. We also did research on what a mic holder really does.

A-C: Defining the Problem

Element A consisted of defining the problem, considering impacts, and doing research. We defined the problem Mr. O'Farrell had for us -- designing microphone holders for the TV Production department. We considered impacts as we elaborated on how new mic holder designs would improve the TV production department and its various news shows. We did research by searching the internet for previous mic holders and how microphones work. We saw no challenges except for finding resources on the internet to back up our research and claims. In Element B, considering strengths and weaknesses of the past to solve the problem, we analyzed previous designs of mic holders, and compared and contrasted them with our new design. It was difficult as we had to scour the internet for previous microphone holders. There were not many mic holder assemblies so it was difficult to find a pre-existing model. In Element C, determining requirements for a design solution to be successful, we created a list of priorities ranging from high to low. We defined the priorities, and went into detail on how we would adapt them in our design. Overall, C was easy as all we had to do was use the feedback given from Mr. O'Farrell to come up with high and low priority design requirements.

D: Prototype designing and Ideation

Element D, generating and comparing ideas, was an important phase as it was where we designed our first prototypes for the mic holder. We made 5 sketches, then picked the best

Name: _____

Date: _____

Class: _____

one. We then expanded further on the design we chose. D was a lengthy process as it was used throughout the entire design process. Every CAD design we made was uploaded to element D, as well as our sketches and drawings. The Gantt chart was easy to make, but it was a slight challenge to maintain the schedule of the chart.

G: Prototype building

Element G, bringing the design to life, was the actual development stage where we 3D printed our prototypes. We used the Onshape designs from document D and our school's Prusa 3D printer to make the physical prototype. This stage was a lengthy process as it was the documentation of all of our 3D prints. We made 5 different prints, each revising on the ones before it. We started by making our holder assembly be fixed with screw on threads, but when that print repeatedly failed we switched to using velcro. Our design requirements naturally aligned with the prints, and each design was a straightforward process on Onshape.

H-I: Testing

Element H was the testing phase. We had 5 different tests ranging from high to low priorities. We held numerous different trials of each test to determine accurate data. We also got feedback from field experts verifying our process. Our tests were an overall success with mostly good results. Element I was composed of the representation of our data on a bar graph. We listed the results from the tests and an explanation of quantifiable metrics. In the future, we will make more adept graphs using more complex design softwares.

J: Evaluation

In Element J, we presented the input from stakeholders and field experts. After our Design-A-Thon presentation to our stakeholders, field experts, and teachers, we completed Element J using the feedback we were given from our stakeholders. We imported the grading rubrics that each stakeholder had, as well as transcripts. We achieved the top score on every category on every stakeholder's rubric. In Element J, we also elaborated on the qualifications of our stakeholders.

Lessons Learned About the Engineering Design Process (K3)

After the Engineering Design experience, the process was a very useful tool in developing our product. The process got redundant at times, especially elements A-D, but after

Name: _____

Date: _____

Class: _____

finishing the process they were important parts of the experience. Each step was necessary to develop the product, and every step built upon the last. Engineers should find this process as a helpful tool, and in the future it could possibly be improved upon by adding another required step such as getting feedback from the everyday person and not a professional.